

DTDR–ALSTM: Extracting dynamic time-delays to reconstruct multivariate data for improving attention-based LSTM industrial time series prediction models

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ARTICLE INFO

Article history:

Received 8 July 2020

Received in revised form 18 September 2020

Accepted 6 October 2020

Available online 22 October 2020

Keywords:

Multivariate time series prediction

Dynamic time-delay

Data reconstruction

Attention-based LSTM

Industrial process

ABSTRACT

Taking advantage of varying degrees of attention on specific features, attention-based long short-term memory (ALSTM) networks have made inroads into the industrial multivariate time series prediction sector recently. However, conventional ALSTM models usually employ static time-delays to constant select input/output pairings of multivariate data, which apparently ignores dynamics of transmission time between industrial process variables and degrades the prediction performance by incorrectly extracting process characteristics. In response to this problem, this paper proposes a novel approach to extracting dynamic time-delays to reconstruct (DTDR) multivariate data for an improved ALSTM prediction model. Therein, the temporal locations and spans of multivariate data are adaptively tailored to input/output pairings of the ALSTM network according to the dynamic time-delays. Specifically, the multivariate data can be accurately matched in temporal positions, and the data information in the original temporal spans with Status transfer time abnormal are replaced. Consequently, this prediction model not only appropriately utilizes dynamics between the predicting and correlated variables, but also makes better attentions on key features extracted from optimum data. Applied to industrial distillation and methanol production processes, the proposed method demonstrates the capability of significantly improving network training speeds as well as prediction accuracies in contrast to static time-delay based ALSTM and LSTM models, expecting even more applications.

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1. Introduction

Taking advantage of modern industrial information technologies, industrial processes have accumulated large amounts of multivariate time series data which contain rich information and knowledge associated with product qualities, operational performances, as well as operating experiences [1]. Additionally, changing trends of multivariate time series can represent dynamic evolving behaviors of multivariable processes together with multilevel structural characteristics [2]. Therefore, it makes sense that multivariate time series predictions of industrial processes can provide valuable supports for process monitoring, fault diagnosis, operational decision-making, etc., which have received widespread concerns in both academia and industry ever since [3].

As a popular data-driven method, neural networks are of powerful nonlinear processing and superior generalization capabilities, which are proved most potentially effective in complex

industrial system predictions [4]. Because of the static structure hard to extract temporal information, feed-forward neural networks are rarely employed in multivariate time series predictions. Alternatively, recurrent neural networks (RNNs) introduce local or global feedback connections in forward signal paths, which allows taking into account temporal information of data so as to achieve better memories and dynamics [5]. However, RNNs usually suffer severe gradient disappearance problems in using backpropagation and gradient descent algorithms, which seriously discourages the use in practice. In response, the long short-term memory (LSTM) is developed. The LSTM establishes a long time lag among the input, the feedback, and the increasing gradient which forces internal states in a special memory unit to maintain a continuous error flow, preventing gradient bursting or disappearing [6]. In addition, the LSTM allows information to selectively affect each moment state in the network through some specified gates, saving important data information for a long time thereby greatly improving the ability to predict multivariate time series [7].

Usually, time series data associated with complex industrial processes are characterized by high dimensions, time-delays and nonlinearities. With the increasing demand for accurate process

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predictions, many scholars have performed extensive studies on conventional LSTM models along with input multivariate data processing. To improve the LSTM structure, Cho et al. proposed a Gated Recurrent Unit (GRU) model characterized by the internal structure with the reset gate and the update gate to control the flow of data information, further improving the network performance [8]. Subsequently, in order to cooperate with the sewage treatment, Zhang et al. used the GRU model to predict water level of the Combined Sewer Overflow structure in Norwegian cities [9]. Ma et al. proposed a Bi-Directional LSTM neural network which consists of a forward and backward LSTM to extract temporal features in two directions, effectively capturing the periodicity and regularity of urban travel flow data [10]. It is found that existing network models usually pair multivariate data with fixed temporal spans during learning and predicting [11]. Wang et al. used principal component analysis (PCA) to screen LSTM network input variables to reduce the data dimension, which enables the prediction of Wind Turbine-Grid Interaction more accurately [12]. Zhou used an improved gray correlation algorithm based on similarity and proximity to select water quality time series before establishing a water quality prediction model based on LSTM [13]. Guo used optimal static time-delays to adjust the temporal spans in multivariate time series pairings [14]. In order to better deal with effects of time-delays and overfitting problems, Zhou et al. proposed a deep LSTM neural network and used static time-delays to select input information of the network, which was applied to the prediction of air quality indicators in a certain area [15]. Yang et al. proposed an attention-based LSTM (ALSTM) model based on Bayesian optimization to predict the day-ahead PV power output [16].

In industrial processes, there are always time-delays available between correlated process variables. Additionally, the time-delays are usually time-varying due to process dynamics and disturbances. Therefore, the time-delay analysis among multiple process variables becomes essential for most industrial process monitoring relevance [17–19]. Shen employed a correlation coefficient of time-delay Cross-Correlation Analysis (CCA) to identify and extract the time-delay between two non-stationary time series [20]. Shah et al. used the cross-correlation function to perform correlation analysis on process alarms, obtaining transmission time between industrial multivariate data according to the maximum correlation coefficient [21]. However, these methods are usually concerned with estimating time-delays from an overall perspective, rarely further focusing on dynamic characteristics of time-delays over a period of time. In response to this problem, Yang et al. presented an idea of dynamic time-delay estimations between correlated variables, introducing a method based on the elastic sliding similarity window to calculate the time-delay at each time point between process variables [22,23].

With the strong dynamic learning ability, ALSTM networks are eligible for processing industrial process time series after appropriate considerations of dynamics of process transmissions. Traditionally, dynamic influences among multivariate data are rarely considered during network training and working, or roughly assigned with fixed static time-delays. As results, multivariate data for network learning are usually specified with constant temporal locations and spans, leading to inappropriate knowledge extractions of industrial process dynamics and deteriorating the prediction performance. Motivated by these observations, this paper proposes a novel approach exploiting the dynamic time-delay to reconstruct (DTDR) multivariate data for improved ALSTM (DTDR-ALSTM) prediction model. The DTDR is essentially a reconstruction algorithm for selected multivariate data during network learning and working. The novel idea behind DTDR includes three major steps: (a) the corresponding temporal locations are adjusted and the contextual abnormal data within the original

temporal spans are replaced according to the distributions and change extents of the historical time-delays between relevant variables and predictor; (b) updated temporal locations and spans of multivariate data are tailored to the input/output dynamic pairings of the ALSTM model learning; and (c) the time-delay at the next prediction moment are estimated to adaptively select multivariate data for network working.

In conclusion, main contributions of this study are addressed as follows:

- A novel DTDR method for industrial process data reconstruction is proposed, which helps make the corresponding form of multivariate time series conform to the actual causal transmission relationship of the dynamic time scales.

- A DTDR-ALSTM prediction model is accordingly created, which can adaptively select optimum multivariate data pairings and make the attention mechanism more effective.

- Applying to industrial distillation columns and methanol production processes, higher prediction performances in contrast to those using static time-delay based ALSTM and LSTM models have been achieved. To the best of our knowledge, this is the first time that the dynamic time-delays of industrial multivariate time series are considered in the prediction model of the LSTM category.

The paper is arranged as follows. Section 2 introduces fundamentals of dynamic time-delays and ALSTM neural networks. In Section 3, the DTDR algorithm is explicitly described before presenting metrics of exploiting DTDR algorithm to improve deep ALSTM networks. In Section 4, the DTDR-ALSTM approach is applied to multivariate time series predictions of two process industry cases, demonstrating the superiority of the contribution in contrast to ALSTM networks with traditional data structures and LSTM networks. The research work is summarized in Section 5.

2. Preliminaries

2.1. Dynamic time-delays

In industrial processes, multivariate correlations are always available while the interactions are often dynamic. Consequently, the time-delays between correlated variables usually reveal dynamic characteristics. Fig. 1 demonstrates the static time-delay and the dynamic time-delay, where, TD represents the time-delay between two variables, *Time* represents the sampling time. It is clearly seen that the static and dynamic time-delays are significantly different.

Considering that most industrial processes are of multivariate, nonlinear and dynamic characteristics, a dynamic time-delay analysis with elastic windows (e-DTA) is developed to extract dynamic time-delays. This algorithm is an improvement of traditional cross-correlation function (CCF) methods, which calculates dynamic time-delays between variables based on sliding elastic time windows and similarity correlation analysis.

The similarity analysis is a popular approach used in time series mining, which statistically calculates the similarity between time series so as to determine the distance or time-delay information between time series. Considering that $\mathbf{X}$ and $\mathbf{Y}$ are two defined time series, $\mathbf{U}_x$ and $\mathbf{U}_y$ are averages of N observations, $\mathbf{R}_x$ and $\mathbf{R}_y$ are standard deviations, and the overall time-delay is m , the similarity correlation coefficient between time series can be characterized by

$$P_{xy}(k) = \frac{E[(X_i - U_x)(Y_{i+k} - U_y)]}{R_x R_y}, \quad m = -n + 1, \dots, n - 1. \quad (1)$$

In the case of the largest absolute value of P_{xy} , the corresponding m becomes the static time-delay between the time series.

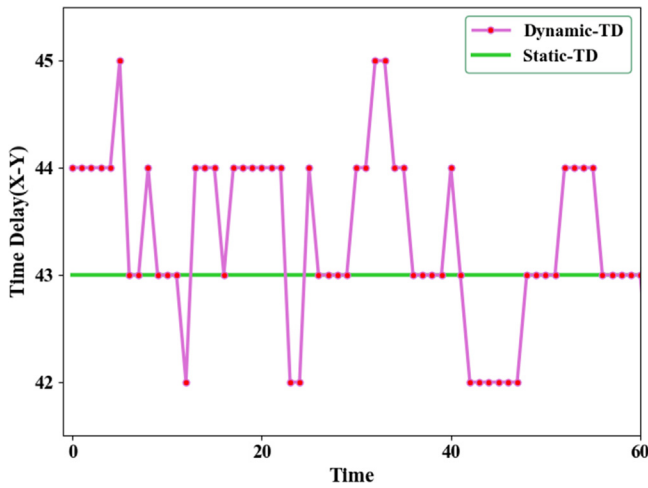


Fig. 1. Schematic diagram of the static and dynamic time-delays.

Static time-delay information can provide supports for subsequent dynamic time-delay extractions. Considering that S_x and S_y are standard deviations of the sample, the similarity coefficient of the data segment is obtained by calculating the expectation of the sample, shown as follows.

$$\hat{P}_{xy(i)}(k_i) = \frac{1}{n_k} \sum_{j=i}^{i+n_k} \frac{(X_{j-r} - U_{x(j)}) (Y_{j+k-r} - U_{y(j+k)})}{S_{x(j)} S_{y(j+k)}}, \quad (2)$$

$$i = 0, 1, 2, \dots, n, k = 0, 1, 2, \dots, \lambda_k.$$

where, the sliding width λ_k is the moving range of the smooth window, and i represents the time point. Setting the observation width n_k as the size of the smoothing window, r as the observation radius of the observation window with $r = n_k/2$, the moving range of the window is determined after judging the current data trend and process conditions empirically. Generally, we have $\lambda_k = W_1 * m$, where $W_1 \in [1, 8]$ is the empirical coefficient, and m is the static time-delay corresponding to the overall data segment. The schematic diagram of time window sliding is shown in Fig. 2.

$P \in [-1, 1]$ is used as the correlation coefficient between variables, whose positive and negative values reveal the positive and negative impacts of the correlation between variables. By observing the maximum value $P_{xy(i)}^{\max}$ and the minimum value $P_{xy(i)}^{\min}$ of the correlation coefficient, the time-delay $\lambda(i)$ of the

variables X and Y can be determined by following equations.

$$\lambda(i) = \begin{cases} k_i^{\max}, & \text{if } P_{xy(i)}^{\max} \geq -P_{xy(i)}^{\min} \\ k_i^{\min}, & \text{if } P_{xy(i)}^{\max} < -P_{xy(i)}^{\min} \end{cases} \quad (3)$$

A major concern of the dynamic time-delay extraction algorithm is the size of the observation window n_k , extensions of which should reference the following observations. (1) Whether there is a too large similarity of a reverse time-delay due to the local strong interference (positive overall time-delay directions) (2) Whether the absolute value of the correlation coefficient between concerned variable data is greater than that of the variables under the overall data. Considering the overall correlation coefficient of the current data, the sliding width λ_k , and prior knowledge, a relatively small value is assigned as the initial value of n_k by trial and error to facilitate subsequent expansions. The algorithmic flowchart is shown in Fig. 3.

2.2. Attention-based LSTM neural networks

Recently, deep learning networks have achieved an unprecedented prosperity from both theoretical and technical perspectives [24]. As a special type of RNNs that can learn long-term dependent information, ALSTM networks have made considerable breakthroughs in many issues related to the time series analysis [25,26]. The ALSTM features the introduction of attention mechanisms in traditional LSTM prediction models, which pays more attentions to key parts of the input sequence affecting output results, allocating more attentions to the key influencing factors as well as better utilizing information from the input sequence. Obviously, it reveals more advantages in dealing with industrial multivariate time series [27].

Specifically, hidden layers of LSTM networks are deepened and independent computing areas of multivariate variables are added with respective attention weights, so that overall feature information is integrated, as shown in Fig. 4.

LSTM neurons are recognized as the core of the network model, shown in Fig. 5. It can be found that, compared to the internal neurons of traditional RNNs, LSTM neurons are of the ability to remove or add information to the next state of a neuron through a carefully designed structure called a 'gate'. The gate is a way to let information pass through, which includes a sigmoid layer σ determining how much information can be passed together with a multiplication operation. The forget gate, input gate, and update gate are located from left to right.

Referring back to Fig. 5, the workflow of LSTM neurons mainly consists of following steps.

Step 1: h_{t-1} represents the output of the previous moment, and X_t is the data input at the current moment. After the forget

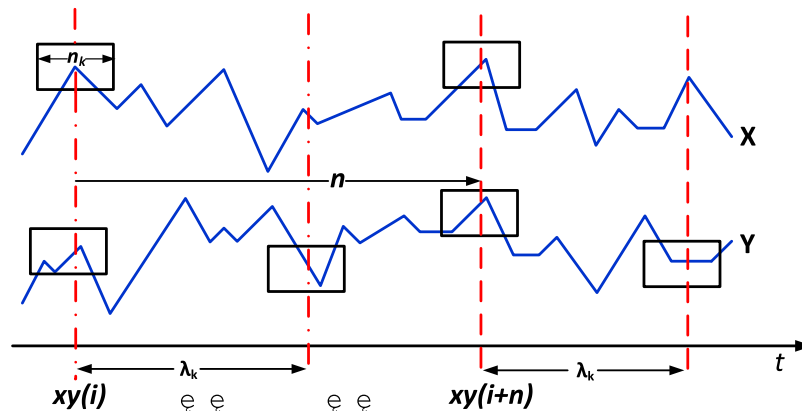


Fig. 2. Sliding time windows.

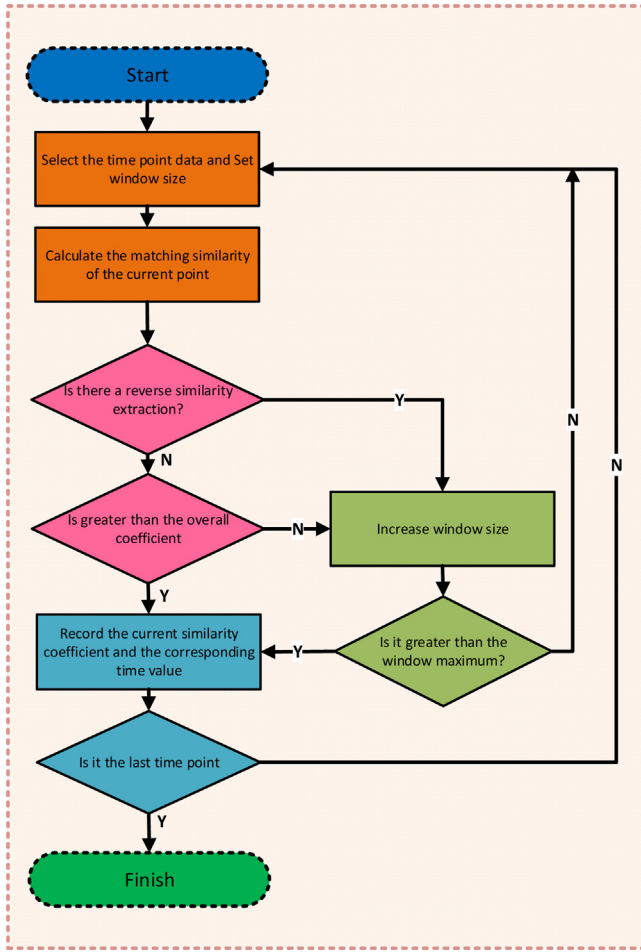


Fig. 3. The e-DTA algorithm extracting dynamic time-delays.

gate which decides what information is discarded, f_t is expressed as follows.

$$f_t = \sigma(W_f \bullet [h_t, X_t] + b_f) \quad (4)$$

Step 2: At this time, we should determine what kind of information the neuron needs. That is, input information is filtered

through the input gate. i_t corresponds to data information to be retained, and a new candidate vector u_t is created and added to the overall status update. The expression is presented as follows.

$$i_t = \sigma(W_i \bullet [h_{t-1}, X_t] + b_i) \quad (5)$$

$$u_t = \tanh(W_u \bullet [h_{t-1}, X_t] + b_u) \quad (6)$$

Step 3: Update the old state C_{t-1} of the neuron to be the new state C_t . Multiply the old state with f_t to discard needless information before adding new information. In time-series prediction applications, such work can be considered as discarding needless information and adding new information based on the predicted goal. The updating expression is presented as follows.

$$C_t = f_t * C_{t-1} + i_t * u_t \quad (7)$$

Step 4: Determine which part of information needs to output through the output gate before processing the current neuron state C_t through $\tanh$, finally multiplying the two results. The specific expression is presented as follows.

$$O_t = \sigma(W_o \bullet [h_{t-1}, X_t] + b_o) \quad (8)$$

$$h_t = O_t * \tanh(C_t) \quad (9)$$

After forming the feature vector $H = \{h_1, h_2, \dots, h_t\}$ through the hidden layer of LSTM, the data information of multivariate data enter the attention mechanism. This behavior will make the network model adaptively focus on the most significant feature information for the current predicted value, which will play an important role in the application of multivariate time series prediction. The attention mechanism will look for the attention weight α_i of h_i , which can be calculated as shown in Eqs. (10) and (11).

$$e_i = \tanh(W_h h_i + b_h) \quad e_i \in [-1, 1] \quad (10)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{i=1}^t \exp(e_i)}, \quad \sum_{i=1}^t \alpha_i = 1 \quad (11)$$

where W_h is the weight matrix of h_i . The new features $H^* = \{h_1^*, h_2^*, \dots, h_t^*\}$ generated by the attention mechanism depend on the multiplication of α_i and h_i , which can be described as Eq. (12).

$$h_t^* = \alpha_i * h_t \quad (12)$$

Finally, the features information of the multivariate data are integrated through the fully connected layer to obtain the predicted value. We use back-propagation through time (BPTT) for

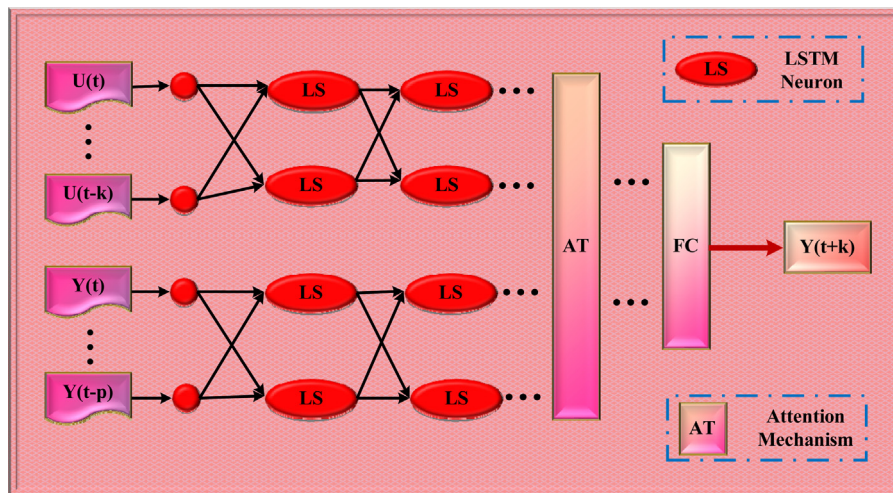


Fig. 4. Deep ALSTM neural networks. $U(t)$: the historical data of related variables, $Y(t)$: the historical data of predicted variable, LS: LSTM neuron, AT: the attention mechanism, FC: the fully connected layer, $Y(t+k)$: predicted value.

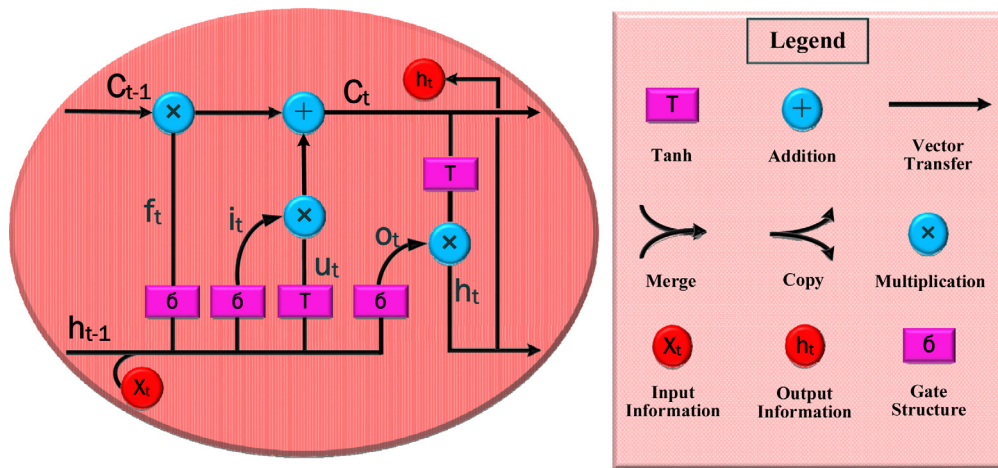


Fig. 5. Internal structures of LSTM neurons.

ALSTM network training, including attention mechanism, which is widely used as an effective training algorithm [28,29].

3. The DTDR- ALSTM model

In current ALSTM applications to industrial time series predictions, static time-delays are usually employed to constantly select multivariate data in terms of temporal locations and spans during network learning and working, which might cause the network incorrectly learning or even missing crucial process information, resulting in prediction performance deteriorations. Taking into account of the dynamic impact of time-delays, a novel approach to exploiting dynamic time-delays to reconstruct (DTDR) multivariate data for the improved ALSTM prediction model is proposed accordingly. Therein, multivariate data used for network learning and working are dynamically reconstructed in terms of constantly updated temporal locations and spans concerning dynamic time-delays. Here, we introduce how to specify temporal locations and spans for multivariate time series before presenting the specific DTDR-ALSTM network models, consequently creating the overall multivariate time series prediction method.

Here we highlight the main differences between our model and recent novel RNN frameworks [30,31] of time series predictions. (1) Our model emphasizes reconstructing the time locations and spans of the originally disordered industrial multivariate data at the initial level to provide effective data knowledge for the ALSTM network. Taking advantage of optimizations of the network model structure, the performance of general models can be certainly improved, but the accuracy of basic data knowledge is the key concern of ameliorating networks. For example, a deep learning model that includes the transformation mechanism [32] is an excellent method to solve the mutation problem of multivariate time series, but applications for industrial processes requires effective data knowledge transfer at the basic level to better utilize the network advantages. (2) Our model focuses on applications in the process industry and financial fields et al. Because the multivariate time series produced have causalities with dynamic time scales in these fields. General forecasting models based on structural optimization are expected to have wide applications. (3) Because that the proposed model utilizes prior knowledge of causality transmitted between the multivariate data, it provides a certain degree of interpretability for the RNNs prediction framework. This behavior is of important engineering theoretical significance, which is rarely provided by existing models.

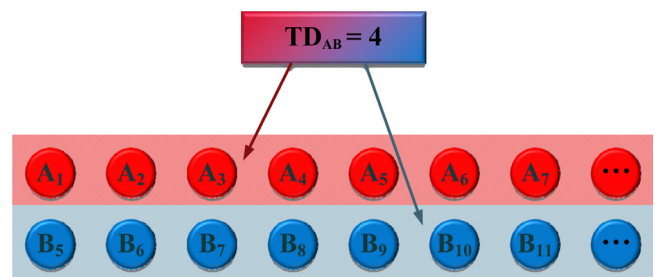


Fig. 6. Temporal locations based on static time-delays.

3.1. Multivariate time series reconstructions

The proposed DTDR is essentially a reconstruction algorithm for selected multivariate data during network prediction. The novel idea behind DTDR is tailored the temporal locations and spans of selected multivariate data for each predicted value according to the distributions and change extents of dynamic time-delays.

The temporal location can be considered as the time point where multivariate variables start to correlate with each other. In this context, temporal locations of multivariate data affecting the predicted time series can be determined directly by time-delays among the multivariate data pairings. Considering time series data associated with two variables **A** and **B**. The time-delay characteristics are extracted through simulations of several sampling time points. Fig. 6 shows the time locations for multivariate time series based on static time-delays. It is seen that all data points of variables **A** and **B** locate with a fixed time-delay. According to the positional relationship expressed by the static time-delay of the overall data as shown in Fig. 6, most traditional algorithms select the corresponding data of related variables in a fixed temporal location during network learning and working.

However, in practical industrial processes, transmission time-delays between process variables are dynamic due to the impact of complicated process correlations and disturbances. Because that time-delays are changing with different operating conditions practically, the multivariate time series should be adjusted with the temporal location according to the extracted dynamic time-delays in a piecewise or point correspondence manner, as shown in Fig. 7. This algorithm highlights accurate positional relationship between the variables in terms of the dynamic time-delay, resulting in appropriate temporal locations of input and output time series data employed during network learning and working.

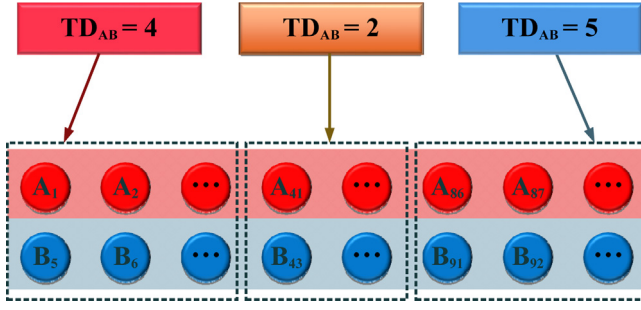


Fig. 7. Temporal locations based on dynamic time-delays.

It is conceivable that time series predictions are susceptible to the selected length of past input data points. Here, the temporal span is considered as the range of past time series data affecting the predicting data. In this context, in order to select the most appropriate multivariate data to accurately and quickly predict the values of variables, most suitable temporal spans in addition to temporal locations are also demanded, which can help reduce redundant data information effectively.

Traditionally, after the determination of temporal locations of selected multivariate data according to static time-delays, temporal spans of the multivariate data are empirically specified. However, due to the shifted process dynamics caused by internal interactions and external disturbances, some outliers associated with the multivariate time series are possibly generated. It makes sense that removing the abnormal data could help improve the prediction performance. Practically, time series data associated with obvious abnormal observations are easily to be removed. However, it is rather difficult to find out contextual abnormal information or useless information of related variables with respect to the predicting data. While considering the difficulty in revealing the abnormal states, the nature behind dynamic time-delays could provide with a novel solution for this problem. During stable process operating periods, it is found that state transmission time between multivariate time series changes little or remains constant. Otherwise, due to the impact of large fluctuations on the process, the transmission time between variables would changes significantly. In this context, it can be concluded that the change extent of dynamic time-delay between time series can reflect shifted process operating conditions to a certain extent, which helps find the best temporal spans of multivariate time series for the predicting ones.

The change extent of relatively reasonable time-delays are described in Eq. (13), where, τ_t represents the time-delay between the output and input multivariate time series at time t , c represents the empirically determined time range concerning time-delays. In addition, in order to better adapt to the dynamic changes of the prediction model, λ_t should be adaptively updated with time. Eq. (14) describes the normal time-delay extent of the data that should be selected, where, τ_{t+k} represents the time-delay between the predicting and correlated variables at the predicting moment.

$$\lambda_t = \frac{1}{c} (|\tau_t - \tau_{t-1}| + |\tau_{t-1} - \tau_{t-2}| \cdots + |\tau_{t-c-1} - \tau_{t-c}|) \quad (13)$$

$$|\tau_{t+k} - \lambda_t| \leq \tau[normal] \leq |\tau_{t+k} + \lambda_t| \quad (14)$$

Consequently, the outliers associated with the multivariate time series are replaced with the help of dynamic time-delay information, achieving optimal temporal spans accommodated to the operating environment. Based on temporal locations and spans, the optimum multivariate data tailored to the predicting time series are adaptively selected during the network learning and working, as shown in Fig. 8.

3.2. Exploiting DTDR algorithms to improve ALSTM prediction model

3.2.1. Dynamic multivariate data selections for network learning

Consider that a prediction task involves multiple time series $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$, where $\mathbf{B}$ is the target variable, $\mathbf{A}$ and $\mathbf{C}$ are the correlated variable, and k is the prediction step. The time series prediction is formulated as follows.

$$B_{t+k} = F_{A,B,C} (A_t, \dots, A_{t-m}, B_t, B_{t-1}, \dots, B_{t-n}, C_t, \dots, C_{t-z}) \quad (15)$$

where, A_t , B_t and C_t are current value of the time series, m , n and z represent the data length of input variables $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$, respectively, B_{t+k} is the predicted target. Due to the causal time-delay between the time series, the practical prediction formulation should be expressed as follows.

$$B_{t+k} = \begin{cases} F_{A,B,C} \begin{pmatrix} A_{t+k-\tau[AB]}, \dots, A_{t+k-\tau[AB]-m}, \\ B_t, B_{t-1}, \dots, B_{t-n}, \\ C_{t+k-\tau[CB]}, \dots, C_{t+k-\tau[CB]-z} \end{pmatrix} & (\tau \geq k) \\ F_{A,B,C} (A_t, \dots, A_{t-m}, B_t, B_{t-1}, \dots, B_{t-n}, C_t, \dots, C_{t-z}) & (\tau < k) \end{cases} \quad (16)$$

where, $\tau[AB]$ and $\tau[CB]$ respectively represent the time-delay of $\mathbf{A}$ and $\mathbf{C}$ time series to $\mathbf{B}$ time series at time point t . Assume that $\mathbf{B}$ time series lag behind those of $\mathbf{A}$ and $\mathbf{C}$, indicating a positive τ . In practical engineering, human operators are generally encouraged to predict key process variables with relative time lags by means of monitoring changing trends of correlated variables. In this context, we are always concerned with the prediction in the case of $\tau \geq k$, in which, the maximum contribution data tailored to appropriate temporal locations and spans could be involved into the prediction formulations, definitely improving the prediction performance.

It is seen that the time-delay directly affects time series prediction in terms of space and time complexity, impacting the efficiency and the performance of network learning and working. It seems inappropriate that conventional approaches employ fixed static time-delays in selecting temporal locations and spans of multivariate data pairings. Taking into account the dynamics of time-delays, dynamic selections of multivariate time series tailored to network learning are shown in Fig. 9, where $a_t = t+k-\tau[AB]_{t+k}$, $c_t = t+k-\tau[CB]_{t+k}$, and del represents the data to be deleted. In so doing, the maximum contribution data set with $A_{t+k-\tau[AB]_{t+k}}$ and $C_{t+k-\tau[CB]_{t+k}}$ as the core that affects B_{t+k} can be utilized. The attention mechanism in the ALSTM model adaptively assigns attention weights to the multiple influencing factors formed by the optimum input data. Therefore, for industrial multivariate time series predictions, the DTDR algorithm can take full advantage of the ALSTM network, reducing redundant data to the greatest extent and ensuring full information for network learning.

3.2.2. Adaptive multivariate data specifications for network predictions

Similar to the analysis of dynamic selection for network learning, dynamic time-delay based adaptive temporal locations and spans of multivariate data for network predictions should be considered.

For the predicting time series, time-delays between multivariate variables at future moments have to be estimated online. The dynamic time-delays can be also considered as time series. In this way, different from the e-DTA method used for extracting dynamic time-delays from historical time series, the LSTM

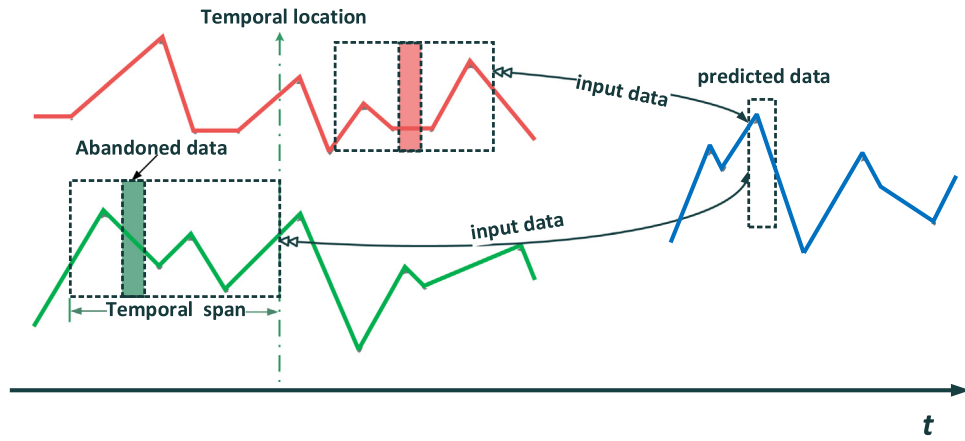


Fig. 8. The temporal locations and spans of selected multivariate data.

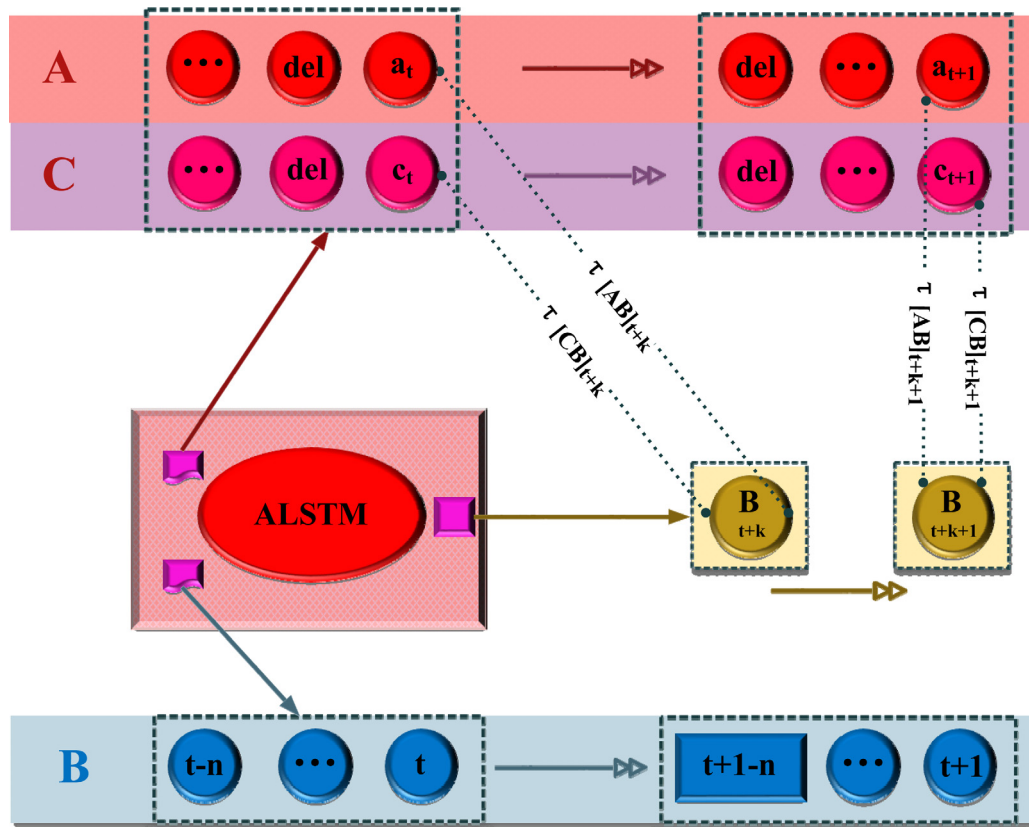


Fig. 9. Dynamic selections of multivariate data for network learning.

network is suggested as the online prediction model of dynamic time-delay sequences, which results from experimental comparisons on multivariate fuzzy [33] interpolation that is widely used and other time-delays prediction methods. As a result, considering **A** and **B** time series, the dynamic time-delay prediction is characterized by

$$\tau[AB]_{t+k} = F_{A,B,\tau}(\tau_t, \tau_{t-1}, \dots, \tau_{t-e}, A_{t-\tau_t}, B_t) \quad (17)$$

where, **e** represents the input data span of the past dynamic time-delay sequence. Thus, according to the estimated time-delay, the corresponding temporal locations and spans can be obtained to implement the online adaptive selection of current multivariate data for network predictions.

3.2.3. The theoretical analysis of the training speed for the DTDR-ALSTM network

In the above sections, the theoretical analysis of improvements of the prediction accuracy is clearly addressed. In addition, the DTDR algorithm can be expected to reduce the computational complexity and accelerate the convergence speed for the ALSTM networks as well.

The DTDR algorithm can reduce the complexity of the multivariate data involved in ALSTM network fitting by reducing redundant data information. Redundant data can be divided into several types. This research is aimed at multivariate data information with abnormal changes of dynamic time-delays. Traditional algorithms rely on static time-delay information indiscriminately to select the time span of input multivariate data. In contrast,

based on the data characteristics represented by the dynamic time-delay, a small amount of contextual abnormal data is discarded, so that the selection time span of the network input becomes smaller and more accurate.

The DTDR algorithm improves the convergence speed by reconstructing the temporal locations of the multivariate data selected for the ALSTM network. Iterative training of neural networks can be regarded as a process of learning nonlinear dynamic characteristics of multivariate data. However, locating the temporal positions of the relevant variable that affects the predicted value based on the disordered static time-delay information will cause the network to make mistakes or omit learning corresponding data knowledge. It is acknowledged that, incorrect and disordered knowledge system transfer will affect the convergence speed of the network model, and correct causality identification will make it easier to achieve convergence. If there is no support from the historical data of the predictor variables, such as some prediction model of the U to Y, the network model based on the static time-delay may not even achieve the expected convergence effect. Because the DTDR algorithm selects the temporal locations of the multivariate data based on the real dynamic time-delays information, characteristics between multiple variables that the network learns can be accurately transmitted correspondingly. Therefore, the DTDR algorithm makes the ALSTM network to effectively exploit the causal transfer relationship of the dynamic time scale between the multivariate time series, thereby speeding up the convergence process.

In summary, in the context of industrial multivariate time series forecasting, the proposed algorithm makes the ALSTM network total number of learning iterations d less and reduce the data complexity $O(t)$ of each iteration, describing as Eq. (18).

$$T_{\text{tradition}} = \sum_{i=1}^d O(t) > T_{\text{DTDR}} = \sum_{i=1}^{d-n} [O(t) - O_{\text{CA}}(t)] \quad (18)$$

where, n is the reduced number of iterations, and $O_{\text{CA}}(t)$ are the computational complexity provided by redundant contextual abnormal data. Therefore, compared with the traditional static time-delay related algorithms, the DTDR algorithm improves the training speed of the ALSTM network.

3.3. Industrial multivariate time series predictions

As mentioned in Section 2.2, the ALSTM network is more eligible for processing industrial multivariate time series than convolutional neural networks (CNNs) and general RNNs. We combine the DTDR method and the ALSTM network for multivariate time series predictions of industrial processes. The DTDR-ALSTM algorithm consists of following sequential steps.

Step 1: Export and filter the required multivariate data from the database.

Multivariate process time series data are exported from DCS historical databases in industrial plants, and then gray correlation algorithms were used to filter the relevant variable data for the prediction task.

Step 2: Extract historical dynamic time-delay information between predictive variable and relevant variables.

Step 3: Multivariate data are accommodated in tailored temporal locations and spans according to the distributions and change extents dynamic time-delays.

Step 4: Establish an ALSTM network model and dynamically select historical multivariate data reconstructed for network learning.

An ALSTM network is designed for the prediction task, in which, the number of LSTM layers are determined according to the data complexity. After the predictive step specified, dynamic pairings of multivariate time series tailored to network learning

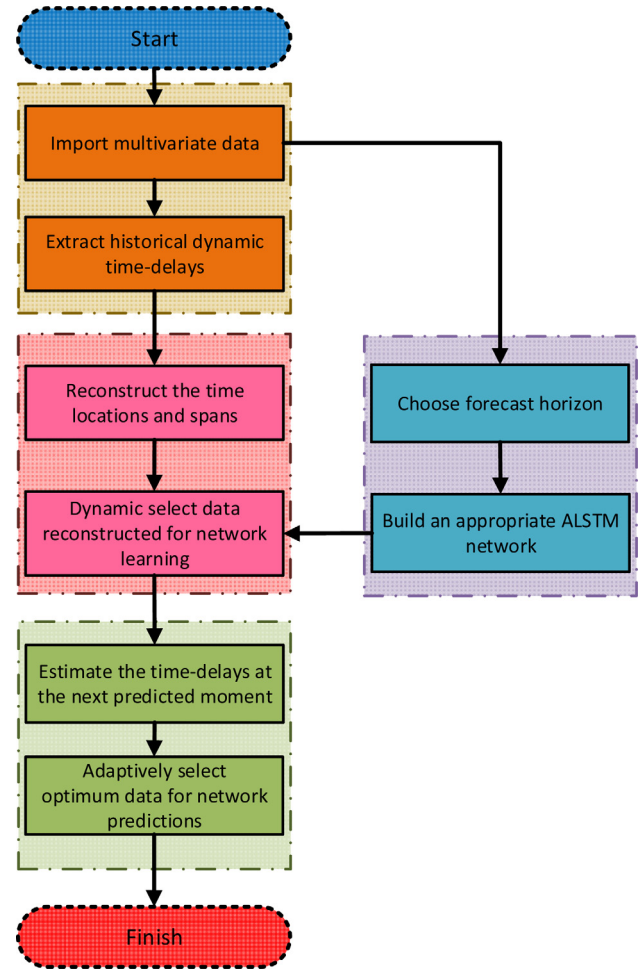


Fig. 10. Flowcharts of the DTDR-ALSTM algorithm.

are carried out. Additionally, multivariate time series data are suggested to be normalized as follows.

$$x_{in}^* = \frac{x_{in} - x_i^{\min}}{x_i^{\max} - x_i^{\min}} \quad (19)$$

where, $x_i^{\max}$ and $x_i^{\min}$ correspond to the maximum and minimum values of the i th time series, x_{in}^* is the normalized value.

Step 5: Estimate the time-delays between the variables at the next predicted moment to obtain corresponding temporal locations and spans.

Step 6: Adaptively select multivariate data based on specified temporal location and span for network predictions.

Consequently, the schematic of the DTDR-ALSTM algorithm is shown in Fig. 10.

4. Case studies

In this section, two industrial examples are employed to verify the proposed algorithm. In Section 4.1, the overall application of the DTDR-ALSTM prediction model is presented in detail. In addition, the experimental details and analysis of the prediction effects of different models are specifically demonstrated. In the application of methanol synthesis process in Section 4.2, we extensively focus on utilizing hypothetical experiments and overall prediction results for data supports and industrial theoretical analyses, comprehensively verifying the feasibility of the model.

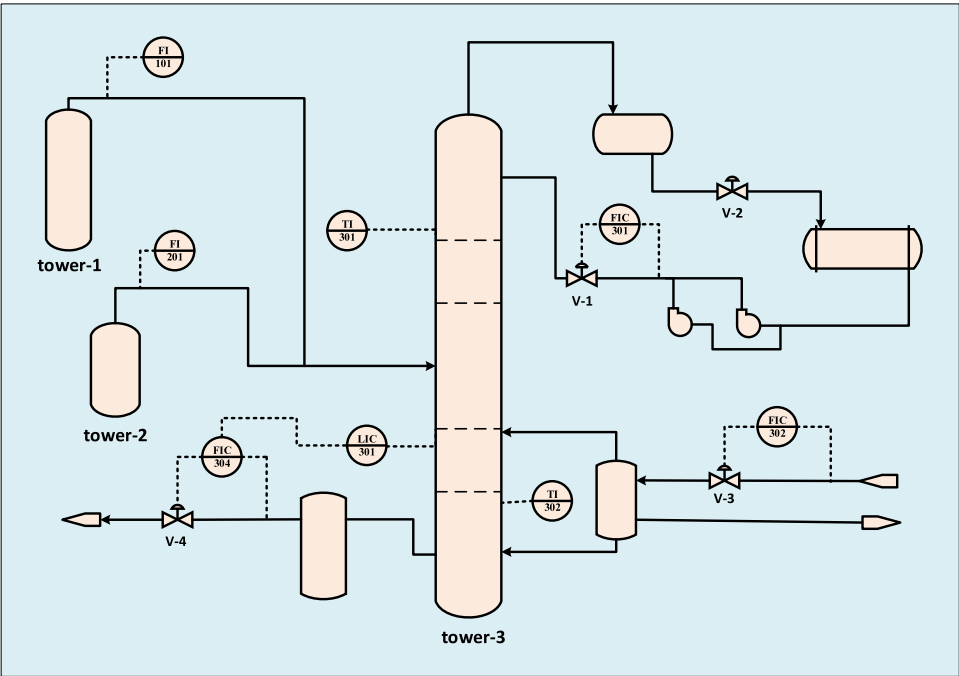


Fig. 11. Schematics of the distillation column plant.

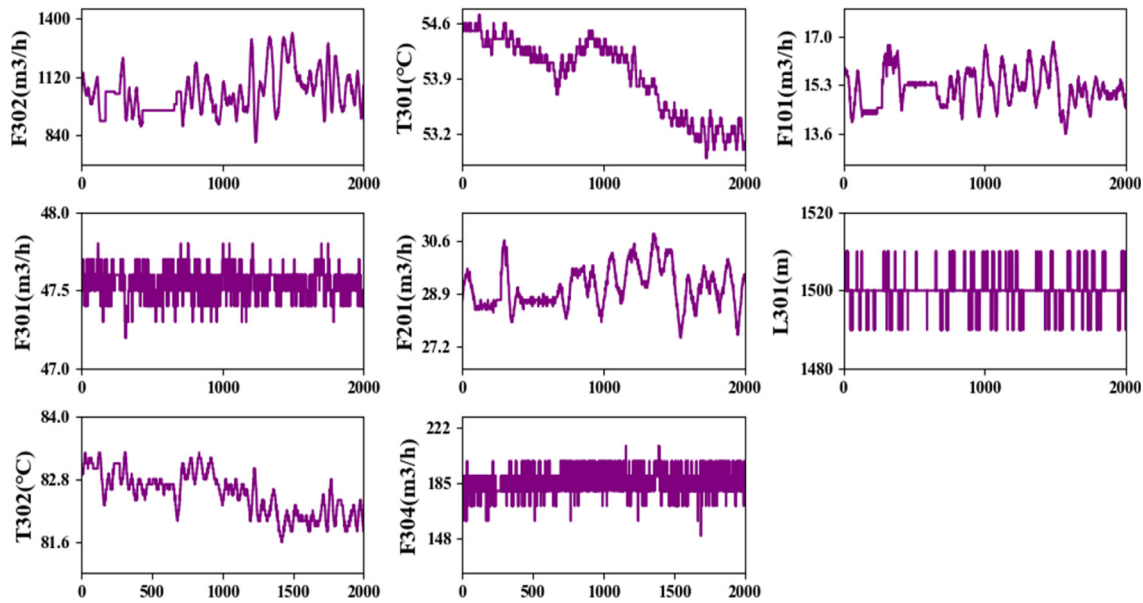


Fig. 12. Distributions of historical data.

4.1. A distillation industrial process

Applications of the proposed approach are carried out using practical data collected from an industrial distillation column plant shown in Fig. 11. Regarding the distillation columns, bottom temperature is recognized as a key process variable, which is usually shifted due to the interaction of process variables and the fluctuation of heat sources. Therefore, multivariate time series predictions of tower bottom temperature are of great significance for the separation and purification of the material mixture.

Consider the major facility, tower-3, along with a top reflux device and a top reboiler device. The main process variables are shown in Table 1, and partial data of the process variables are shown in Fig. 12, where the abscissa indicates the sampling time.

Table 1 Main process variables of distillation.		
Variables	Unit	Description
L301	m	The liquid level of tower bottom
F101	m ³ /h	The top flow rate of tower-1
F201	m ³ /h	The top flow rate of tower-2
F301	m ³ /h	The top reflux flow
F302	m ³ /h	The re-boiling steam flow rate
F304	m ³ /h	The Outlet flow of tower bottom
T301	°C	Top temperature of tower-3
T302	°C	Bottom temperature of tower-3

25,000 time series data associated with the eight process variables in Table 1 were collected, including 15,000 training data

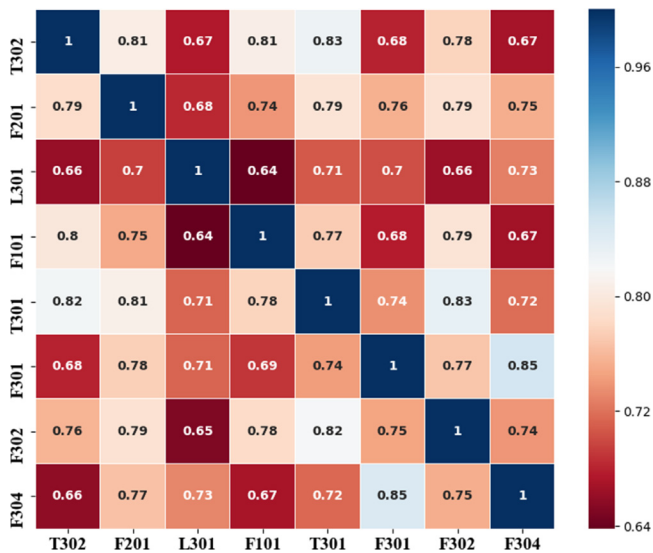


Fig. 13. Gray correlations between variables.

points and 10 000 testing data points, in which, all data associated with multivariate variables are of the same sampling time. The gray correlation algorithm was used to analyze the correlation between bottom temperature and the other seven correlated variables, shown in Fig. 13. A suitable threshold value of 0.7 is chosen to exclude variables F301, F304, and L301. As a result, four strongly correlated variables (T301, F101, F201, F302) and a predicted one (T302) were selected for subsequent studies.

Using the CCF and e-DTA algorithms described in Section 2.1, the static time-delay (STD) and dynamic time-delay (DTD) sequences between the five correlated variables and bottom temperature were extracted from the training data. Fig. 14 shows the STD and DTD sequences at some time points.

Table 2

Prediction accuracy of dynamic time-delays.

Time-delay	Root Mean Square Error (RMSE)				
	t+1	t+2	t+3	t+4	t+5
TD-F302T302	0.1364	0.1856	0.2164	0.2321	0.3138
TD-T301T302	0.1172	0.1417	0.2248	0.2761	0.2951
TD-F101T302	0.1441	0.1942	0.2453	0.3042	0.3662
TD-F201T302	0.0962	0.1542	0.1938	0.2113	0.3051

Taking account of the data complexity, two hidden LSTM layers were specified for the ALSTM network [29]. In order to demonstrate the superiority of the DTDR algorithm over static time-delay (STD) algorithms, the traditional LSTM networks [15] were additionally used as an auxiliary network for comparison experiments. According to practical demands, bottom temperature is predicted from 1 to 5 steps. Based on distributions and change extents in dynamic time-delays between T302 and the four related variables, multivariate data that network learning selected were dynamically reconstructed in term of the tailored temporal locations and spans. All network models are implemented with Keras38 and Theano39 libraries in Python.

During the network training, different methods are compared in terms of learning ability. Taking the prediction step $t + 5$ as an example, the learning curves are shown in Fig. 15. It can be seen that DTDR-ALSTM and DTDR-LSTM models both enjoy better learning abilities, faster training speeds and smaller final training errors over traditional STD-LSTM and STD-ALSTM models, which benefit from more comprehensive and accurate knowledge learned from multivariate time series with the DTDR algorithm.

Subsequently, dynamic time-delays between T302 and related variables are estimated. Accordingly, time-delay predictions are performed with predicting steps 1 to 5. Table 2 presents prediction accuracies, and Fig. 16 shows the prediction performance in the step $t + 5$.

Based on corresponding temporal locations and spans obtained in estimated time-delays information, current multivariate

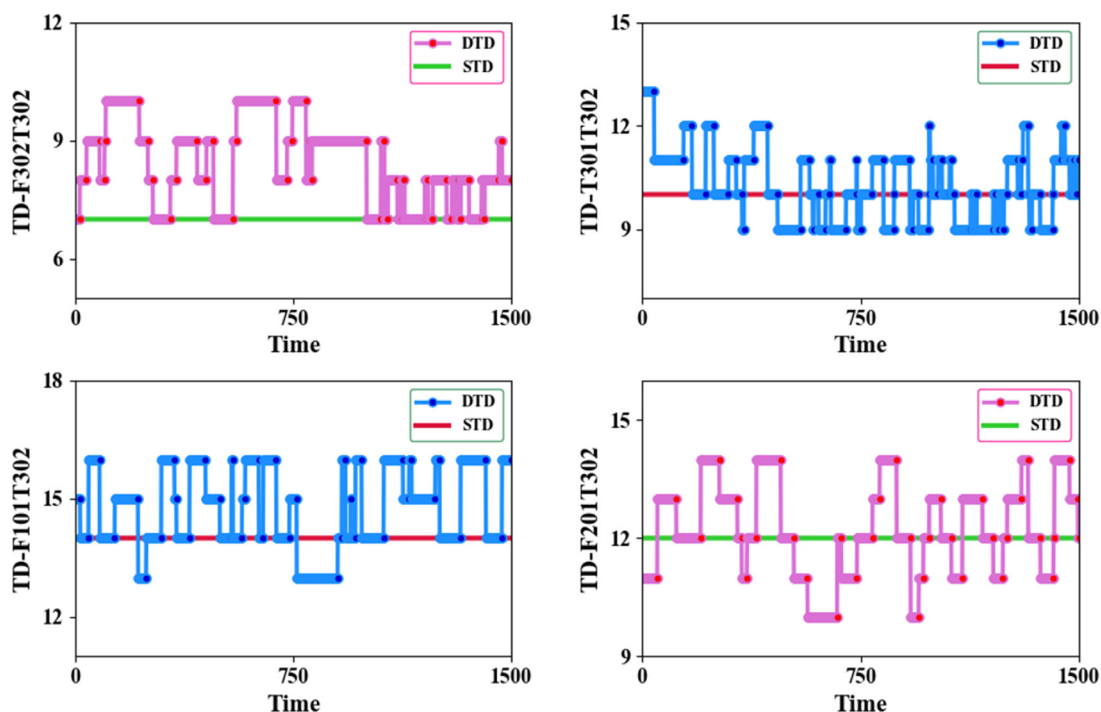


Fig. 14. Distributions of static and dynamic time-delay sequences.

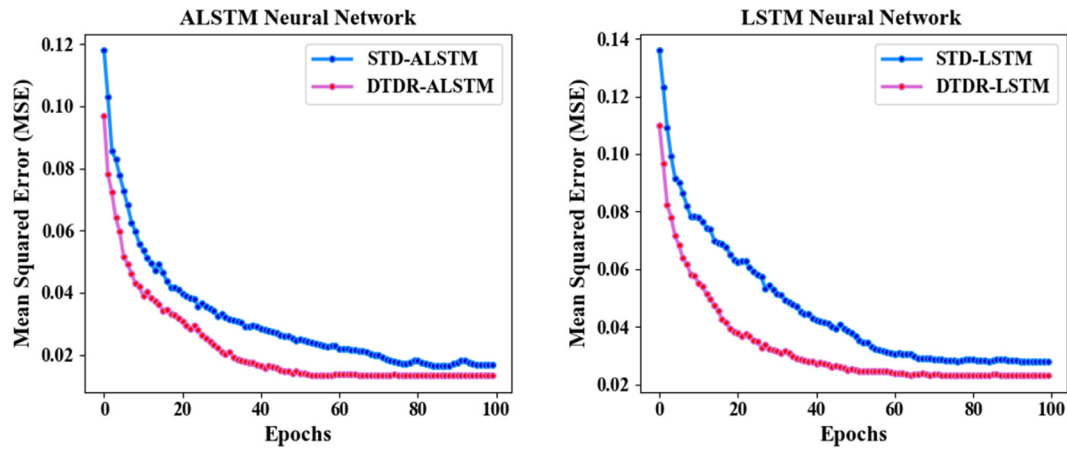
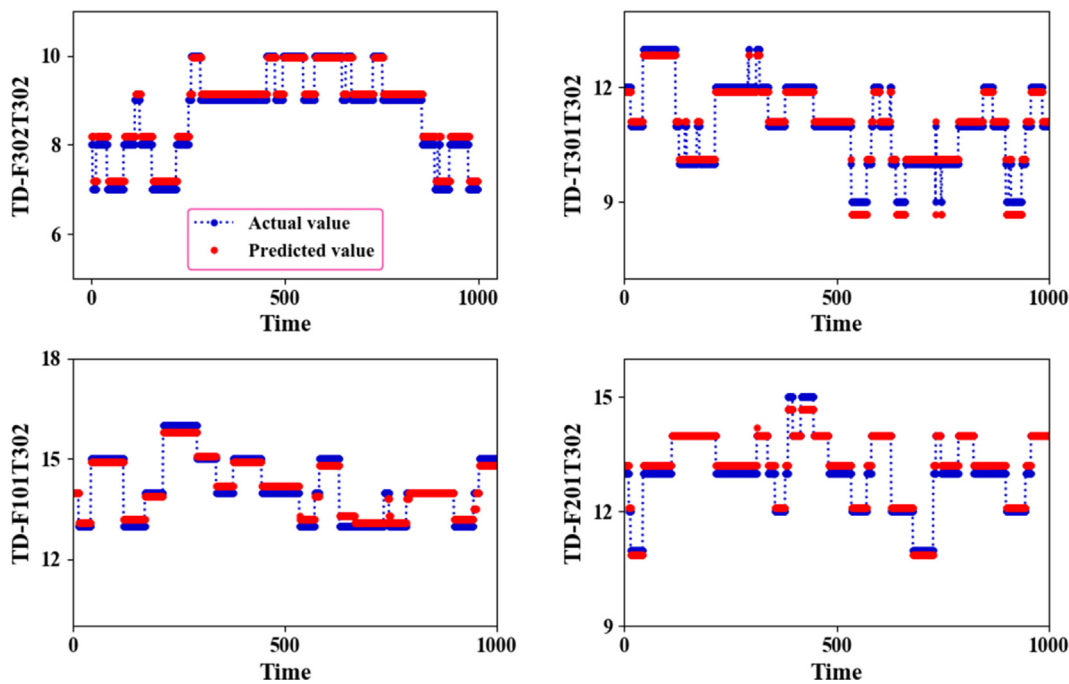


Fig. 15. Learning curves of ALSTM and LSTM networks.

Fig. 16. Estimated results of dynamic time-delays at the $t + 5$ step.

data are adaptively selected for online predictions. The time series prediction of bottom temperature is performed at the step $t + 5$, shown in Fig. 17.

In order to verify the superiority of the proposed algorithm over traditional static time-delay (STD) algorithms, we compared the DTDR-ALSTM method with the traditional STD-ALSTM method, and used the STD-LSTM and DTDR-LSTM methods for additional experimental comparisons. The predictions of T302 at the step $t + 5$ with different methods are shown in Fig. 18 along with the corresponding errors shown in Fig. 19. In the further verifications, two network models are distinguished for comparison. The prediction profiles and errors of T302 at the step $t + 4$ with different network methods are shown in Fig. 20. Fig. 21 shows the prediction results of T320 in different data areas at steps $t + 2$ and $t + 3$.

As two widely used indicators [34,35] in the field of time series predictions, the mean absolute error (MAE) and mean square error (RMSE) are selected to evaluate the prediction results, shown in Table 3. The quantitative indexes of tower bottom temperature

Table 3

Prediction accuracies with different methods from $t+1$ to $t+5$.

Horizons	Loss	LSTM	DTDR-LSTM	ALSTM	DTDR-ALSTM
$t+5$	RMSE	0.152	0.121	0.136	0.094
	MAPE	0.114	0.089	0.093	0.078
$t+4$	RMSE	0.128	0.084	0.097	0.073
	MAPE	0.092	0.071	0.080	0.063
$t+3$	RMSE	0.093	0.063	0.069	0.058
	MAPE	0.084	0.056	0.059	0.047
$t+2$	RMSE	0.072	0.054	0.051	0.045
	MAPE	0.060	0.046	0.043	0.037
$t+1$	RMSE	0.059	0.043	0.039	0.031
	MAPE	0.048	0.035	0.031	0.028

time series prediction at steps 1 up to 5 with different methods are shown in Figs. 22 and 23.

Seen from Figs. 18–21, it is found that the DTDR-ALSTM method has a better prediction performance than those of traditional methods in the multi-step prediction of tower bottom temperature. Referring back to Figs. 22 and 23, it is seen that,

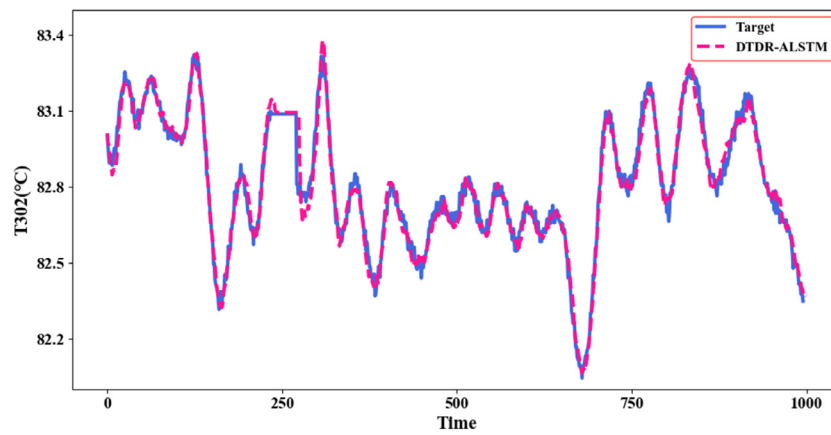


Fig. 17. Prediction results of bottom temperature at the $t + 5$ step.

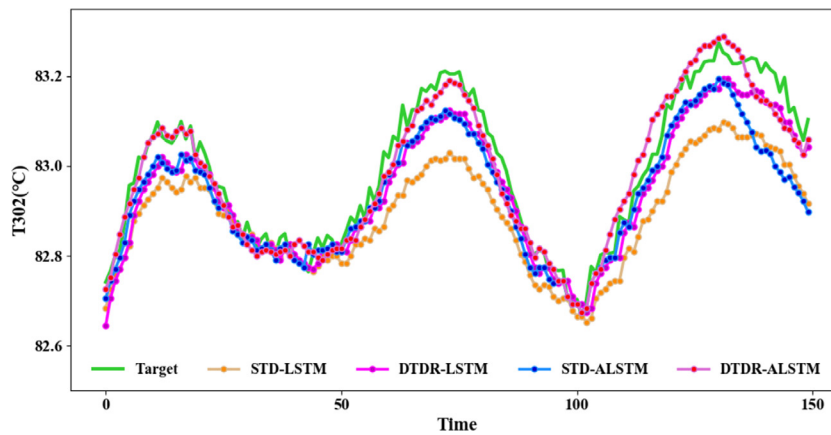


Fig. 18. Predictions results of T302 with different methods at the $t + 5$ step.

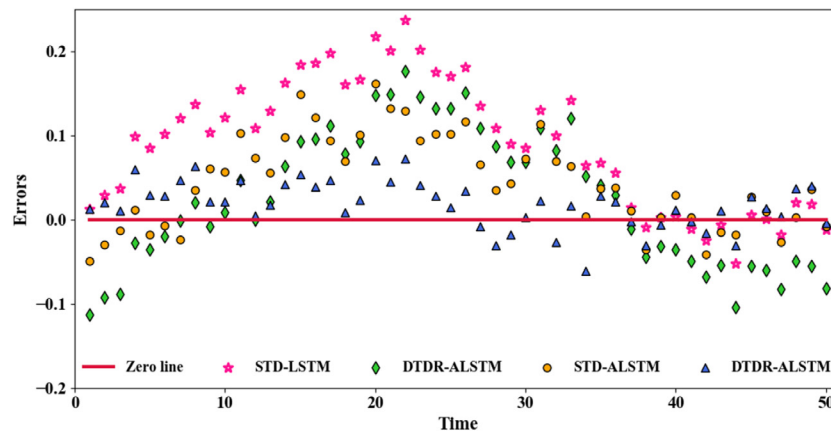


Fig. 19. Prediction errors of T302 with different methods at the $t + 5$ step.

even though the prediction performance of the LSTM network is improved with the help of the DTDR algorithm, it is limited with the traditional structure. However, DTDR-LSTM has smaller prediction errors than those of the STD-ALSTM at steps $t + 5$, $t + 4$, and $t + 3$, which shows that, if the input multivariate data are mismatched in terms of time-delays or invalid, the superiority of attention mechanism on corresponding inaccurate features are greatly reduced. Practically, in the process engineering, shifted predicting variables possibly result from significant changes of relevant process variables in advance. Therefore, appropriate

temporal pairings of multivariate data are selected for the prediction network along with the attentions on key features could definitely improve the predicting performance.

Additionally, seen from the RMSE and MAE in Table 3, it can be clearly found that from $t + 1$ up to $t + 5$, the DTDR-ALSTM approach is of better prediction performance, corresponding to RMSEs of 0.031, 0.045, 0.058, 0.073, and 0.094, respectively. While for traditional STD-ALSTM networks, the prediction accuracies correspond to 0.039, 0.051, 0.069, 0.097, and 0.136, respectively. So, it makes sense to consider that the DTDR-ALSTM algorithm

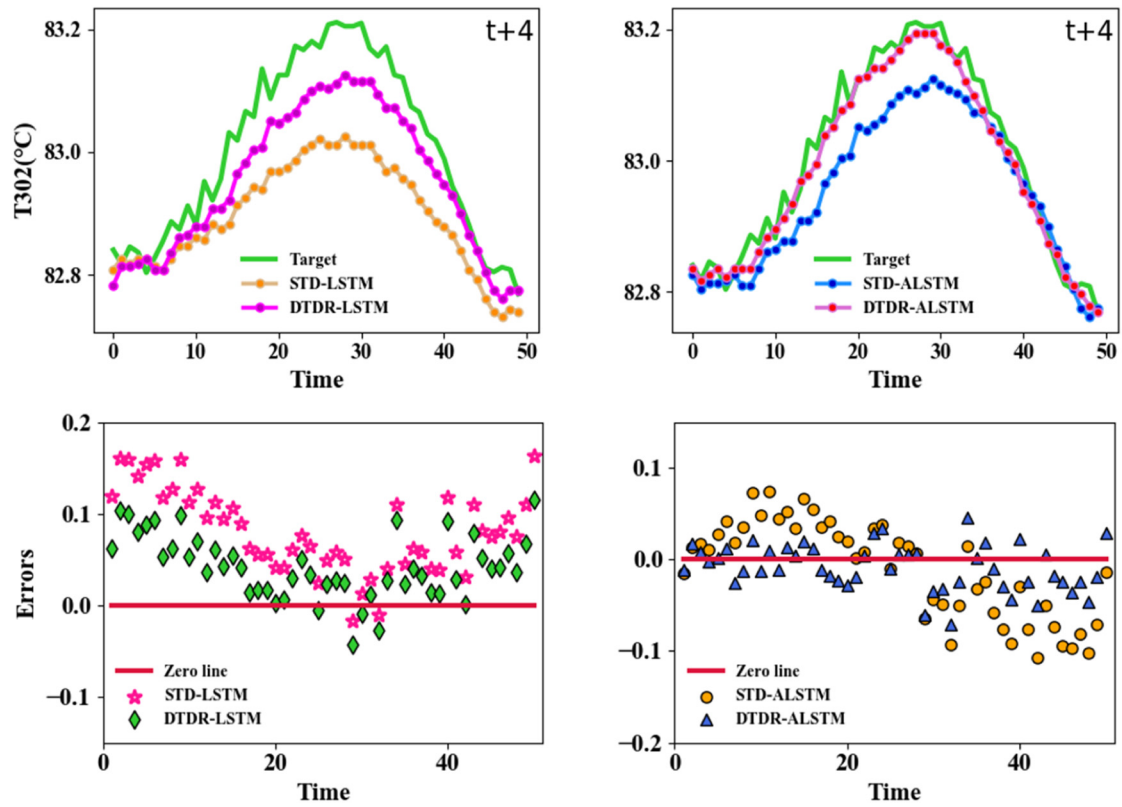


Fig. 20. Prediction results and errors of T302 with different methods at the $t + 4$ step.

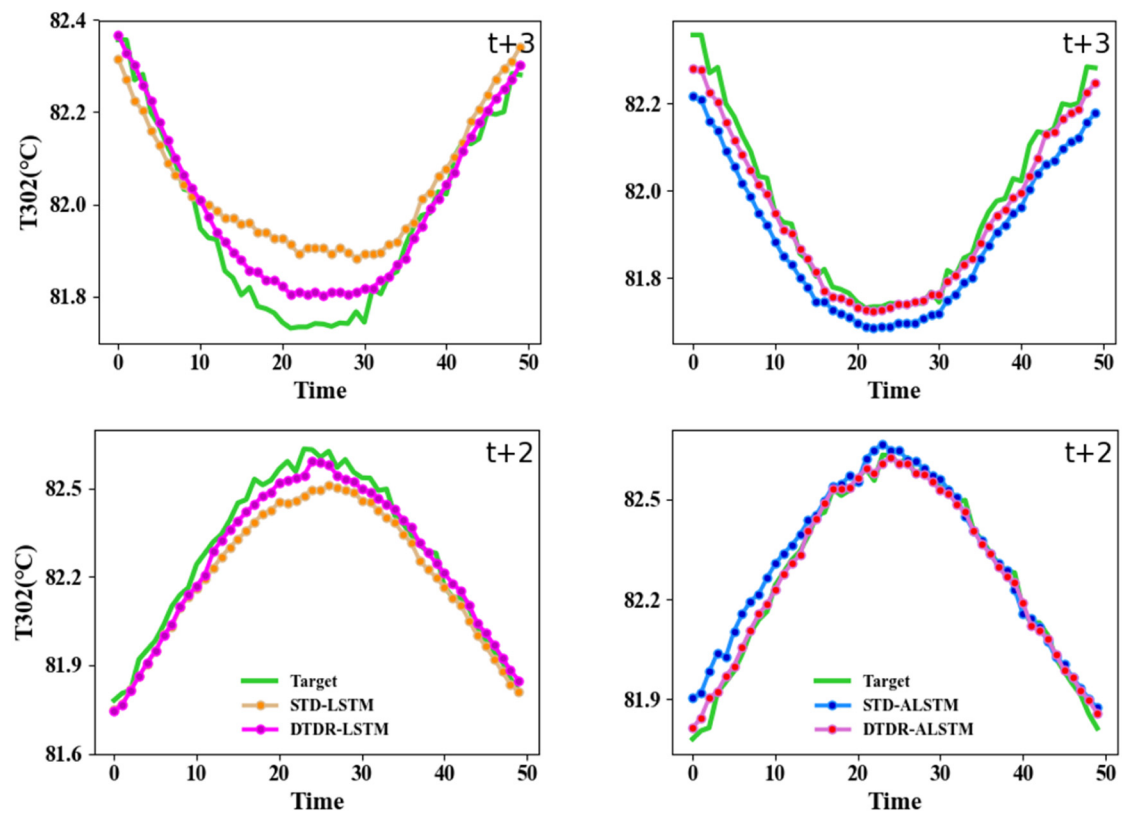


Fig. 21. Prediction results of T302 with different methods at the $t + 3$ and $t + 2$ steps.

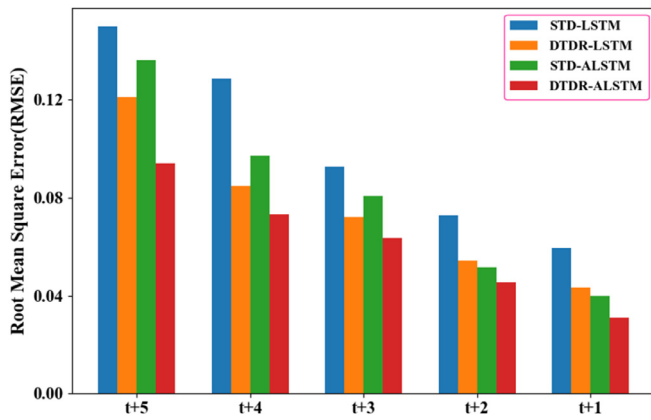


Fig. 22. RMSE values of predictions with different methods.

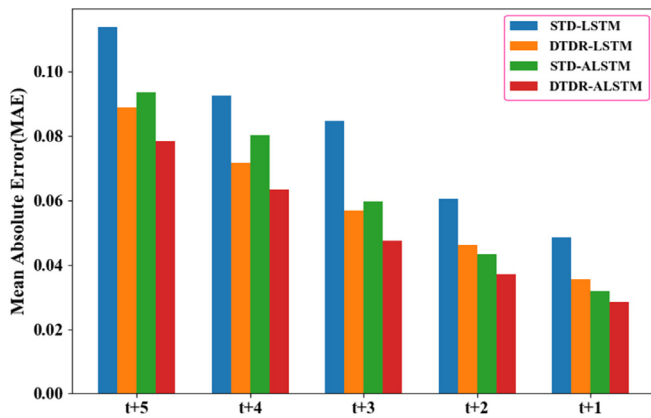


Fig. 23. MAE values of predictions with different methods.

is more suitable for processing multivariate data of industrial processes.

4.2. An industrial methanol production process

Methanol is a kind of green renewable energy with high octane number and low risk [36]. It can be safely manufactured from various carbon-based sources, producing environmentally friendly by-products and providing an artificially manufactured, cost-effective alternative to non-renewable energy. Therefore, many scholars have carried out in depth researches on modeling and optimizations of methanol production process.

Methanol production is considered as a complex multivariate nonlinear process, in which several variables that affect key target variables are involved. In the methanol industrial process, the changing trend of the export crude methanol flow rate is an important indicator reflecting the overall plant conditions and benefits. The operating characteristics of this process lead to the existence of causal transfer relationships of dynamic time scale between multivariate data. In what follows, the methanol process is selected from an industrial plant. Fig. 24 shows the main part of methanol synthesis in the production process. The main process variables are shown in Table 4, and partial data of the process variables are shown in Fig. 25, where the abscissa indicates the sampling time.

A multivariate time series forecast task for crude methanol flow (F3503) was formulated according to actual plant demands. 20,000 sets time series data associated with the 8 process variables in Table 4 were collected, including 15,000 training data

Table 4

Main process variables of methanol synthesis.

Variables	Unit	Description
F3103	N m ³ /h	The airflow of the chemical plant
F3106	N m ³ /h	The hydrogen flow
F3503	t/h	The crude methanol flow
P3302	MPa	Outlet pressure of methanol separator
P3701	MPa	Outlet pressure of methanol releaser
T3010	°C	The temperature of the upper synthesis tower
T3210	°C	The temperature of the lower synthesis tower
T3401	°C	Outlet gas temperature of the upper synthesis tower
T3402	°C	Outlet gas temperature of the lower synthesis tower

points and 5000 testing data points, in which, all data associated with multivariate variables are of the same sampling time. In order to reduce the influence of irrelevant variables on the crude methanol flow (F3503), the process variables in Table 4 were screened using the gray relational analysis algorithm combined with process knowledge. Finally, four main correlation variables (F3103, F3106, P3302, T3402) were specified, which combine with the predicting variable F3503 to construct a group of multivariate time series for the forecasting task.

Subsequently, the CCF and e-DTA methods described in Section 2.1 were used to extract the static time-delays (STD) and dynamic time-delays (DTD) of the four selected correlation variables and the predicting variables, as shown in Fig. 26.

Based on the industrial data, the experiments were performed to verify significant performance improvements of the proposed DTDR-ALSTM prediction model. All network models are programmed using Keras38 and Theano39 libraries in Python.

In the field of multivariate time series, the hypothetical experiments [37] are carried out to verify the feasibility of forecasting models and supports of multivariate data. Using crude methanol flow (F3503), the following tests are performed to exploit the ALSTM networks. Two hypothesis tests are performed as follows.

Test1:

It is assumed that the causal influences between the related industrial multivariate time series are transmitted without time-delays. Therefore, the predictive performance of inputs without relevant variables should be lower than that of inputs with relevant variables. The testing procedure includes:

- (1) using a single variable F3503 for multi-steps forecasting (Y-Y);
- (2) using a method of no time-delay (NTD) to process and select multivariate data for F3503 multi-steps prediction (UY-Y).

The prediction errors of the two models for variable F3503 from $t + 1$ to $t + 5$ steps are shown in Fig. 27. It can be clearly seen that the UY-Y model that contains relevant variable data processed without time-delays cannot improve the prediction performance compared to the Y-Y model, even causing performance degradations. However, according to Grant causality, the input of relevant variables in this technological context should improve the prediction performance. Therefore, it is true that related variables processed without time-delays cannot provide effective data knowledge for the ALSTM network.

Test2:

It is practical that the causalities between the related industrial multivariate time series are transmitted with dynamic time-delays. Therefore, the performance of the prediction model using dynamic time-delay information of multivariate time series should be better than that of using static time-delay information. The testing procedure includes:

- (1) Using the traditional static time-delay algorithm combined with ALSTM (STD-ALSTM) for F3503 corresponding multivariate time series prediction;

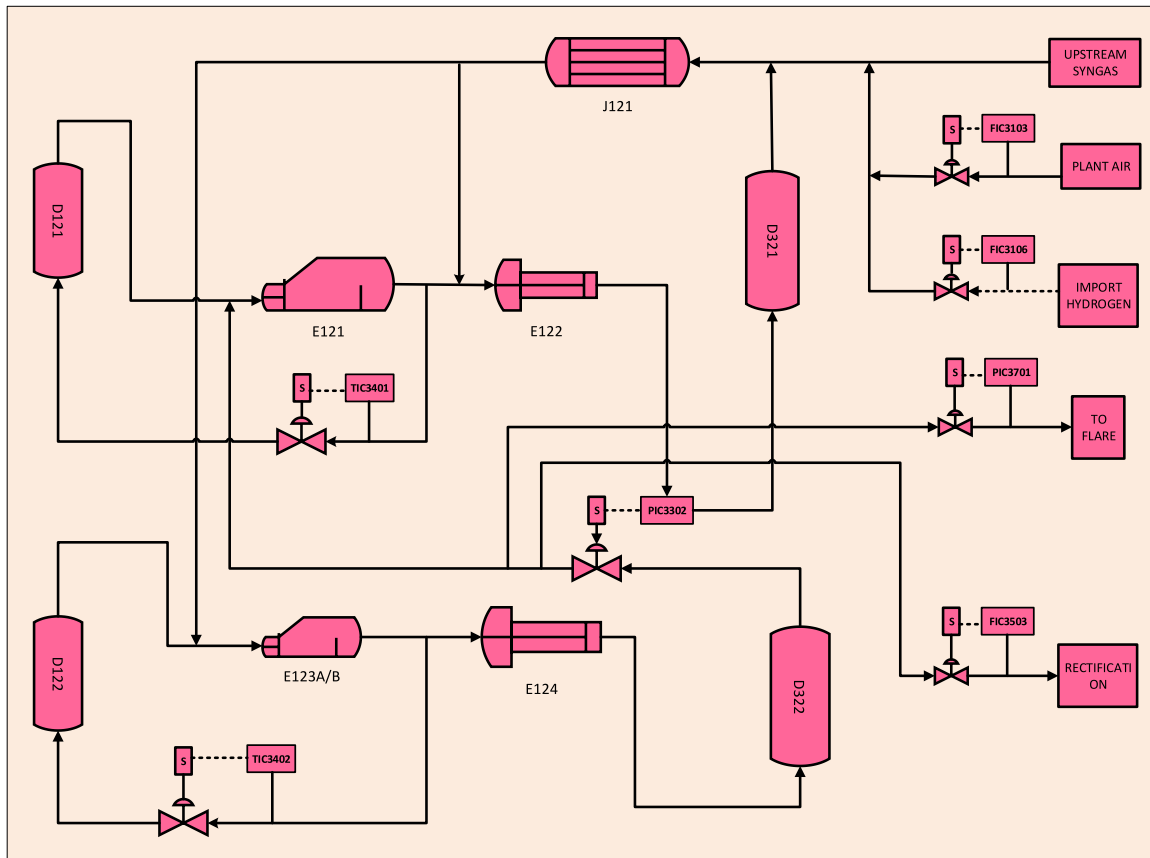


Fig. 24. Main parts in the industrial process of methanol synthesis.

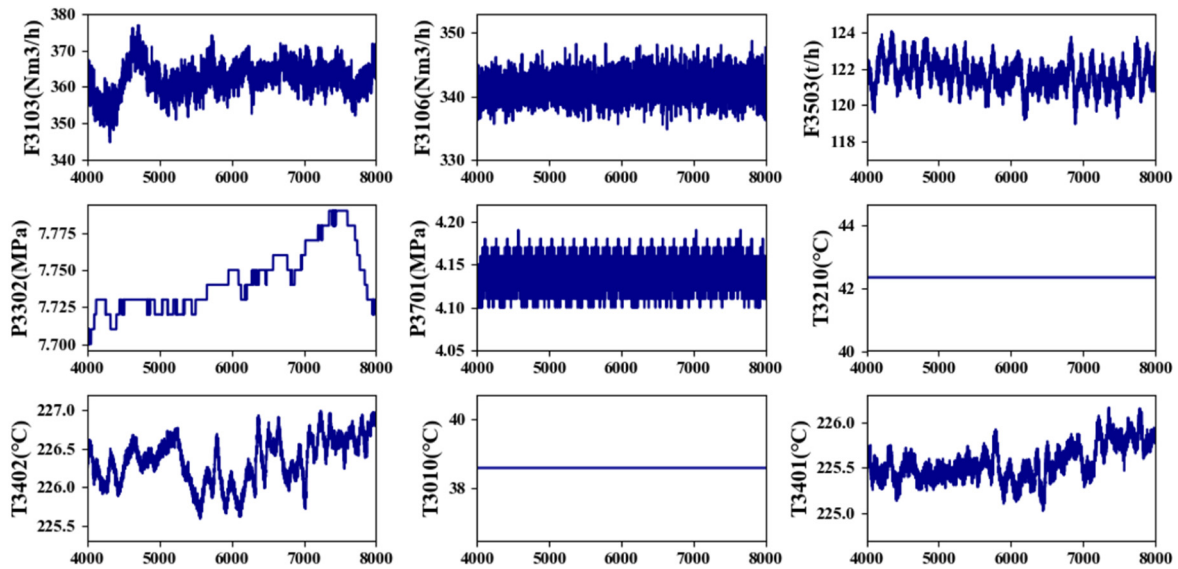


Fig. 25. Distributions of historical multivariate data.

(2) Using the DTDR-ALSTM model for the identical prediction task.

Fig. 28 shows the multi-steps prediction accuracy of the STD-ALSTM model using static time-delays and the DTDR-ALSTM model using dynamic time-delays. It is seen from the comparison results in Fig. 28 that the DTDR algorithm is advantageous than the STD algorithm in terms of improving network prediction performances. The essential reason is that the DTDR can deeply

and accurately extracts the causal relationship characteristics of multiple time series through timing locations and spans reconstruction. So, multivariate time series can accurately tailor to the time dimension and provide effective data supports for the ALSTM networks. This result verifies the hypothesis of test 2 and the irrationality of traditional algorithms that exploit static time-delay (STD) information to transmit knowledge for the ALSTM network.

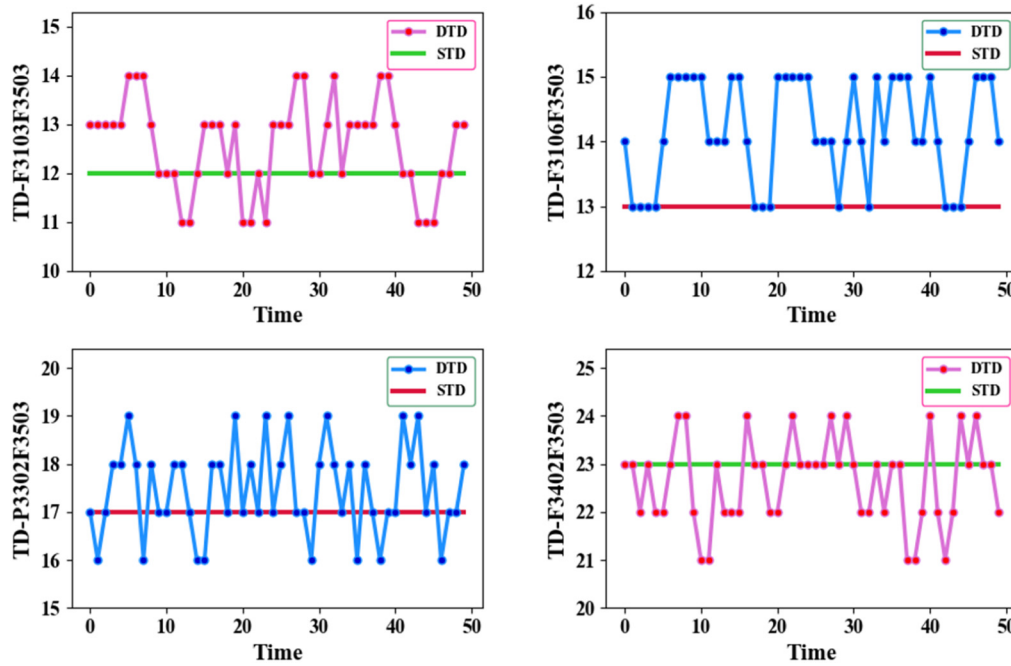


Fig. 26. Distributions of dynamic and static time-delay sequences.

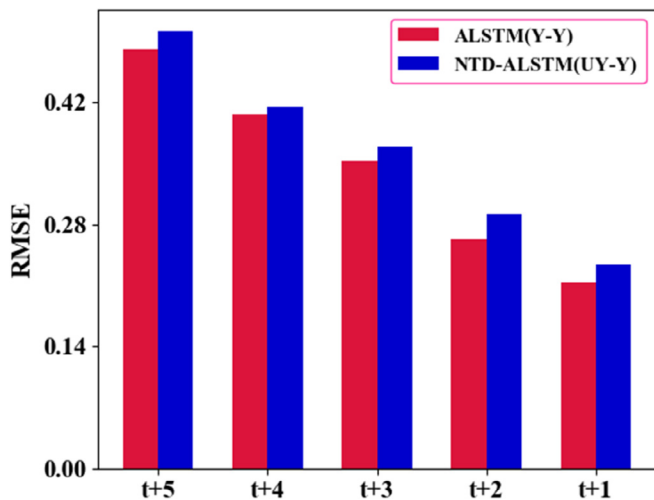


Fig. 27. RMSE values of ALSTM (Y-Y) and NTD-ALSTM(UY-Y).

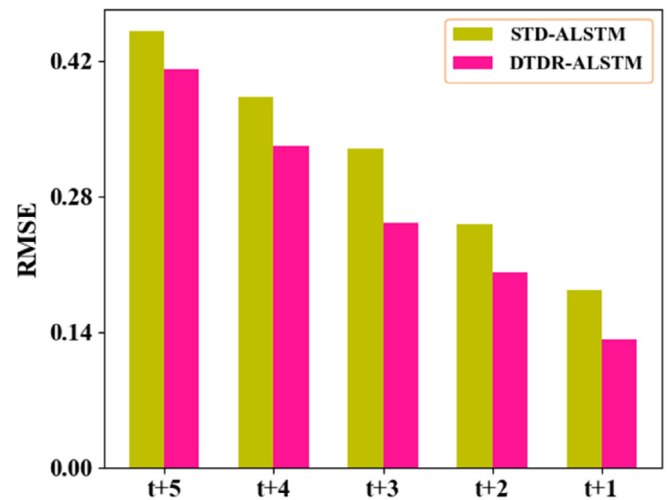


Fig. 28. RMSE values of STD-ALSTM and DTDR-ALSTM prediction models.

Moreover, we synthesize the entire hypothetical experiment and conduct a theoretical analysis. The method accuracy in all hypothetical experiments is listed in Table 5. It is worth noting that the STD-ALSTM achieves a certain performance improvement compared with the NTD-ALSTM and the ALSTM(Y-Y). This result shows that using static time-delay information to manipulate related variables in this process can provide partial industrial data knowledge for the ALSTM model. This is due to the intersection of static and dynamic time-delays form the multivariate time series generated during the methanol processing during this time period. However, it is practical that static time-delays will be greatly different from dynamic ones in many industrial processes according to the normal evolutions of the complex system. In these scenarios, the relevant variable data inputting to the STD-ALSTM will fail to provide effective information, resulting in a sharp decline in prediction performance even close to the NTD-ALSTM model. Because that time-delays in the process industry

are always dynamic, RNNs prediction models using static time-delays possibly even produce errors from the data transmission, inevitably leading to poor overall performances because of the dependency for the quality of multivariate data of deep learning algorithms. Therefore, that exploiting DTDR algorithm truly and accurately extract the causality knowledge between the multivariate time series can improve the accuracy of the ALSTM industrial prediction model from the essence of transferring data knowledge.

Finally, different baseline models are used to compare with the proposed model in learning abilities and prediction performances. Specifically, in the $t + 5$ step of F3503 predictions, the ALSTM [29] and LSTM [15] deep networks combining the dynamic time-delay algorithm DTDR and traditional static time-delays STD method are used to deal with multivariate data, respectively.

Additionally, trial and error methods are used to select the optimal hyperparameters of different network models before the

Table 5
Prediction accuracies with different methods in the hypothesis testing.

Horizons	Loss	Y-Y	UY-U	STD-ALSTM	DTDR-ALSTM
		ALSTM	NTD-ALSTM		
t+5	RMSE	0.482	0.507	0.451	0.411
t+4	RMSE	0.406	0.414	0.382	0.331
t+3	RMSE	0.351	0.368	0.329	0.252
t+2	RMSE	0.262	0.293	0.252	0.203
t+1	RMSE	0.214	0.233	0.183	0.132

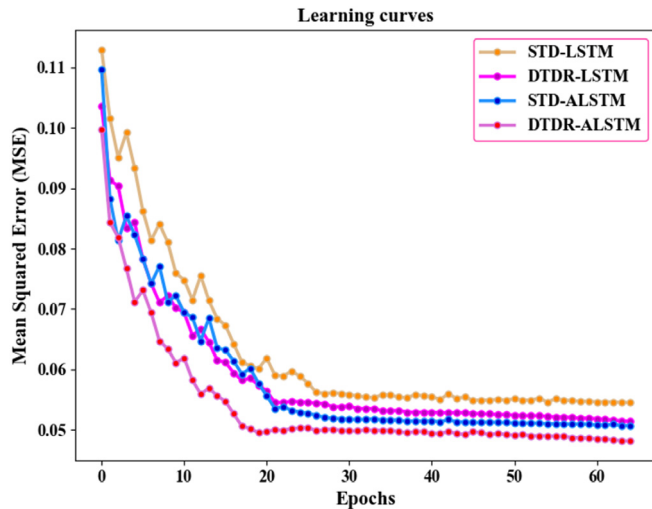


Fig. 29. Learning curves of the different network models.

training processes are carried out. Fig. 29 shows the learning curves of different models. The learning curves show that, because the DTDR-LSTM cannot further break through the limitations of its structure, its learning ability rarely competes with that of the DTDR-ALSTM model. It is seen that the DTDR-ALSTM has excellent learning ability than that of the STD-ALSTM networks, and the DTDR-LSTM and the STD-LSTM have the same conclusion. This is because the time locations of multivariate time series are tailored to the predicting values and able to abandon redundant data that should be selected in the original time span.

The prediction results of different networks for step $t + 5$ of F3503 are shown in Fig. 30. The quantitative indexes of RMSE and MAE values for all prediction steps are shown in Figs. 31 and 32.

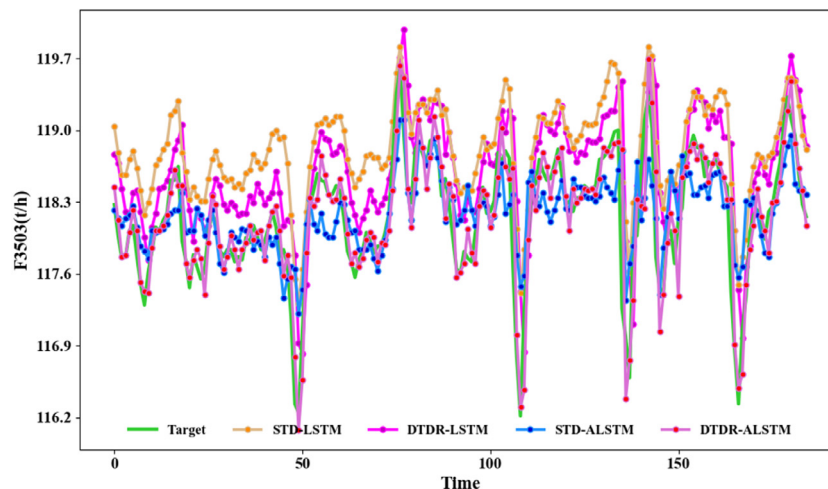


Fig. 30. Prediction results of F3503 with different methods at the $t + 5$ step.

Comparing the prediction performance of DTDR-LSTM with that of the STD-LSTM, the corresponding RMSE values reduce from 0.481, 0.402, 0.349, 0.276 and 0.235 to 0.433, 0.353, 0.309, 0.241 and 0.203, which shows that the DTDR algorithm helps improve the performance of basic LSTM models to a certain extent. It proves that the DTDR algorithm can transform the causality of industrial time series into accurate knowledge at the bottom layer of the deep network for in depth transmission, adapting the constantly updated network structure of RNNs. However, seen from the comparison of the prediction errors of the DTDR-LSTM and the DTDR-ALSTM, it is concluded that the DTDR algorithm cannot break through the limitations of the network structure to make the prediction performance of the basic network model achieving a crucial improvement. Consequently, it reveals that the DTDR-ALSTM model has small prediction errors in all prediction steps than those of other baseline models. For the DTDR-ALSTM model, the corresponding minimum RMSE values are 0.132, 0.203, 0.252, 0.331, 0.411 and MAE values are 0.094, 0.126, 0.175, 0.203, 0.298, which verifies the effectiveness of the proposed model. These error results are also consistent with the actual operation of the process industry. The DTDR-ALSTM model, which exploits data knowledge of causality between industrial multivariate time series effectively, shows excellent predictive performance.

5. Conclusion

The complex nature of industrial processes always leads to multivariate time series data with causal transmissions of dynamic time scale and nonlinear characteristics. In response, a DTDR-ALSTM prediction model based on dynamic time-delay information to reconstruct multivariate data is proposed, which better solves the problem that traditional data structure of networks rarely conform to the dynamic transfer relationship of multiple variables in actual industrial processes. According to the dynamic time-delays, multivariate data are in consistent with the predicting time series in terms of tailored temporal locations and spans. As a result, the network prediction model is able to adaptively select optimum multivariate data during learning and working. This work takes full advantage of the attention mechanism of the network model, which not only reduces the extra redundant information, but also helps to grasp the main points from the process variables, thereby improving the prediction performance. Comparison experiments with different methods have been carried out using time series data from industrial distillation columns and methanol production processes, which

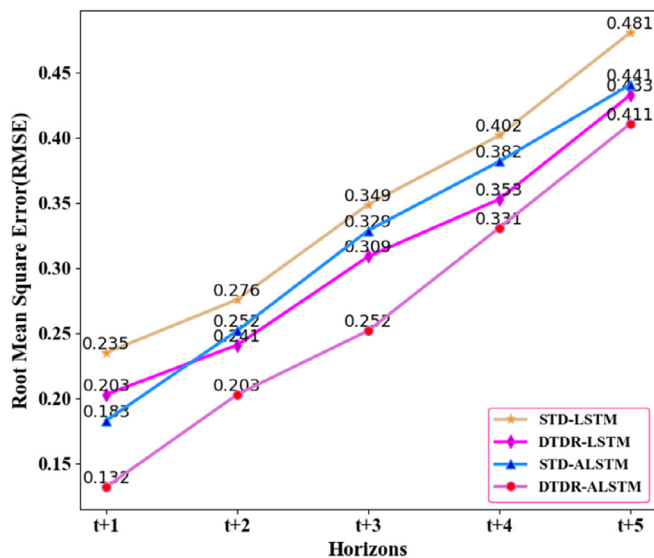


Fig. 31. RMSE values of predictions with different methods from $t + 1$ to $t + 5$.

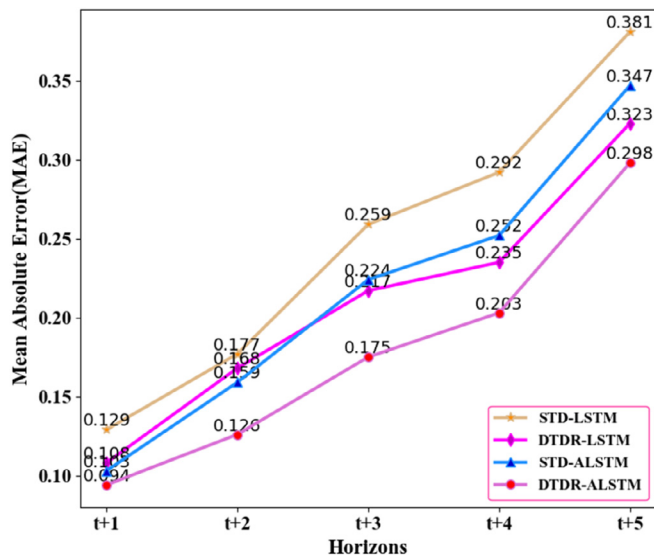


Fig. 32. MAE values of predictions with different methods from $t + 1$ to $t + 5$.

show that the DTDR-ALSTM method can significantly improve the learning speed and prediction accuracy, demonstrating the effectiveness of the novel approach. In addition, the DTDR algorithm exploits prior knowledge in the basic layer of deep network data transmission. As a result, the DTDR algorithm can not only be used in conjunction with the RNN model that is constantly updated at the structural level, but also provides a certain degree of interpretability for the prediction framework.

Our future work will focus on more accurately extracting dynamic time-delays between multivariate time series for industrial processes. In addition, it is expected to apply the proposed method in other fields such as finance with the dynamic time-delay condition.

CRediT authorship contribution statement

Jince Li: Conceptualization, Software, Writing - original draft, Writing - review & editing. **Bo Yang:** Investigation, Supervision,

Writing - review & editing. **Hongguang Li:** Methodology, Formal analysis, Validation. **Yongjian Wang:** Data curation, Supervision. **Chu Qi:** Project administration. **Yi Liu:** Software, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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